



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Human Computer Interaction Final Exam

NAME: Abdul Rafay

ROLL NO: FL23791

PROGRAM: BSSE (5th Semester)

COURSE: HUMAN COMPUTER INTERACTION

SUBMITTED TO: Sir Khateeb khan

DATE: 30 Sep 2025

DAY: Tuesday

Question 01: Designing Icons for “Smart Campus” Mobile Application

Tool Used

Figma

(Icons designed following mobile UI standards)

Icon Sizes

- 24 × 24 px
- 48 × 48 px

Design Consistency

- Same color theme
- Same stroke style
- Simple and minimal design
- Suitable for mobile screens

1. Attendance Icon

Meaning:

Represents student attendance records.

The checkmark/calendar symbol shows presence and tracking of attendance.



2. Time Table Icon

Meaning:

Represents class schedules and daily lectures.

The clock/table layout indicates time management and planning.



3. Fee Payment Icon

Meaning:

Represents payment of university fees.

The currency/card symbol shows financial transactions clearly.



4. Notifications Icon

Meaning:

Represents alerts and important announcements.

The bell icon indicates new updates and messages for students.



HCI Concepts Applied

- **Visibility:** Icons are clearly visible and understandable
- **Affordance:** Icons suggest their function naturally
- **Consistency:** Same style and size across all icons

Question 02: Interactive Cube with Visual Feedback (Unity)

Objective

To demonstrate the HCI principle of **feedback** by changing the cube's color when it moves using keyboard input.

Scene Description

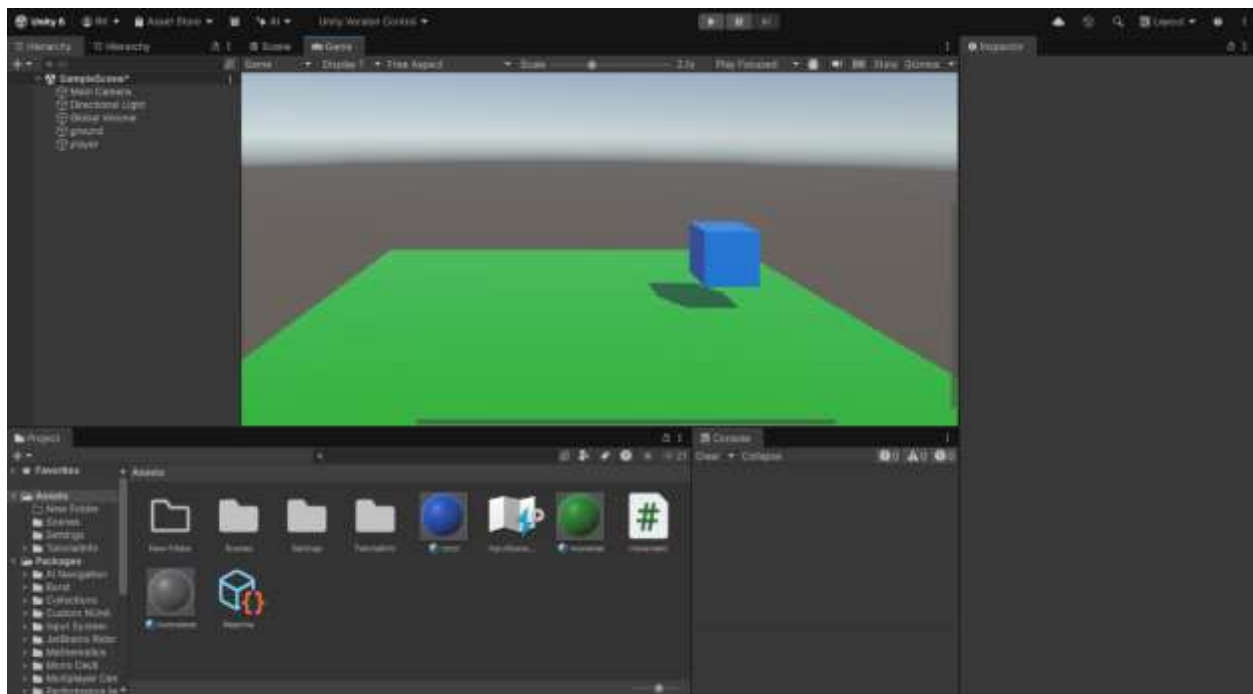
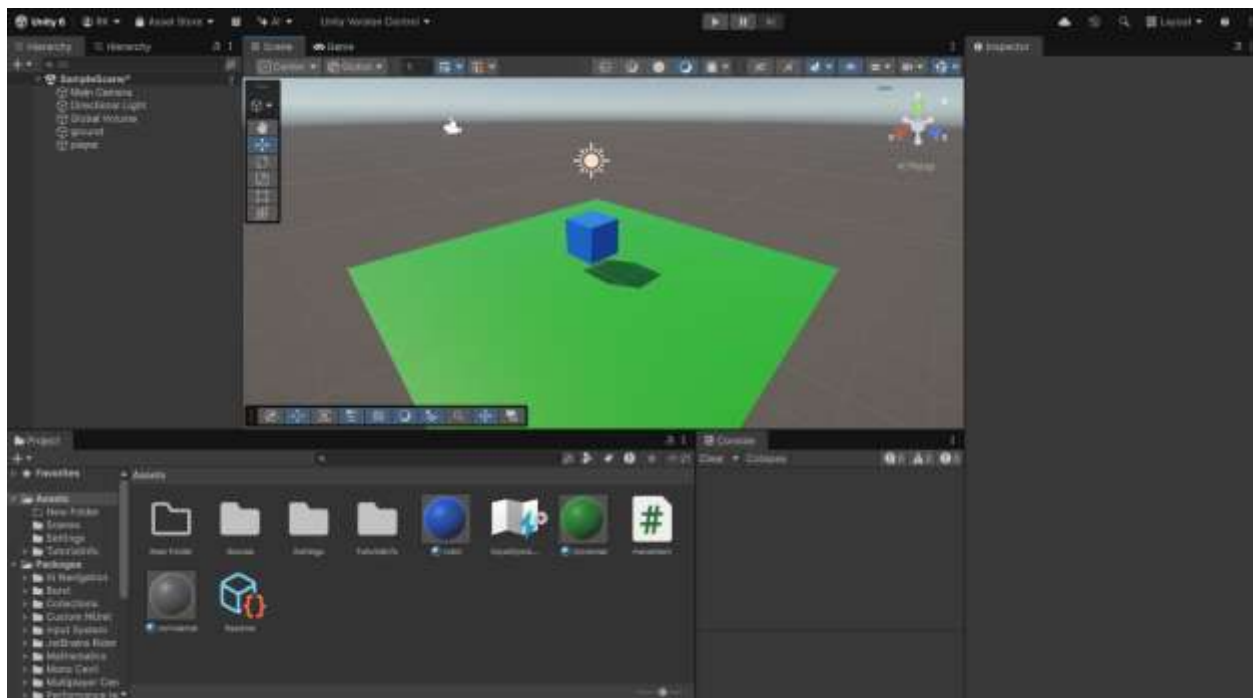
- A **Cube** is placed on a **flat surface**
- The cube moves **left and right** using arrow keys
- Color changes temporarily when movement occurs

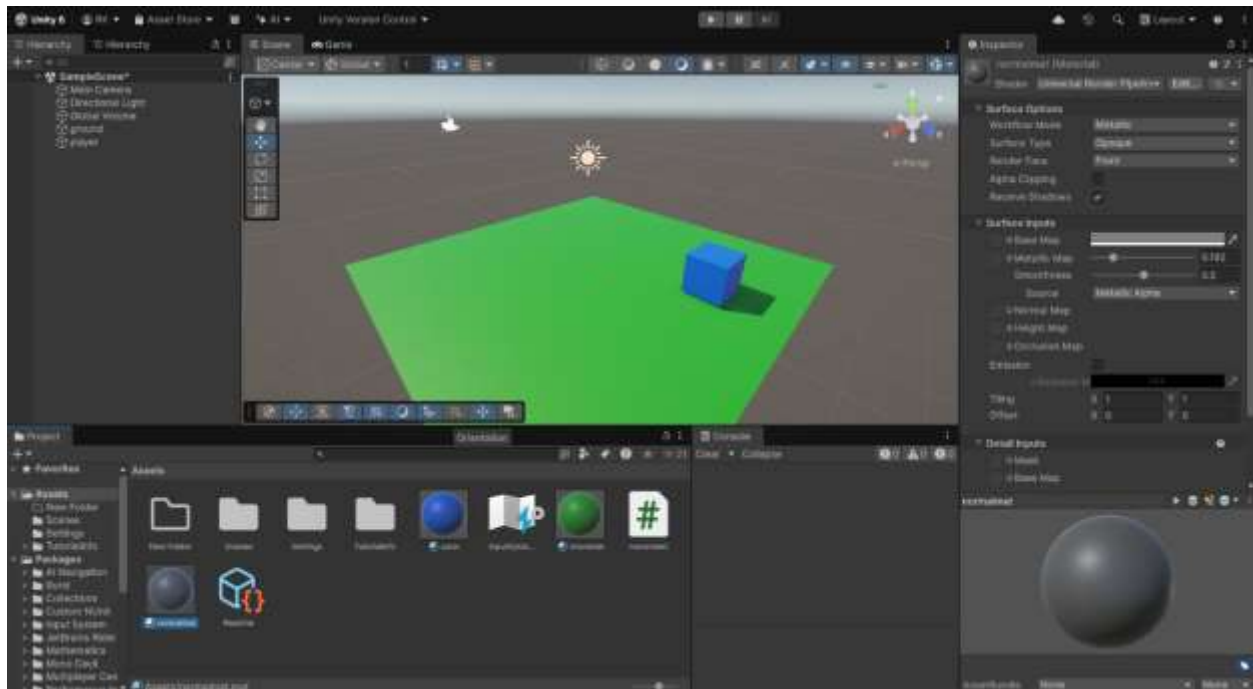
Components Used

- **Cube GameObject**
- **Plane (Ground)**
- **Rigidbody** (for physics)
- **Two Materials** (Normal & Feedback)
- **C# Script** (for movement and color change)

Working of the System

- Left Arrow ← → Cube moves left
- Right Arrow → → Cube moves right
- During movement:
 - Cube color changes to show **visual feedback**
 - After **0.5 seconds**, color returns to normal





Code:

```
using System.Collections;

using UnityEngine;

using UnityEngine.InputSystem;

public class CubeMovement : MonoBehaviour

{

    public float speed = 5f;

    public Material normalMaterial;

    public Material moveMaterial;

    Rigidbody rb;

    Renderer rend;

    void Start()

    {
```

```

        rb = GetComponent<Rigidbody>();

        rend = GetComponent<Renderer>();

        rend.material = normalMaterial;
    }

```

```

void Update()
{
    if (Keyboard.current.leftArrowKey.isPressed)
    {
        rb.linearVelocity = new Vector3(-speed, rb.linearVelocity.y, 0);

        StartCoroutine(ChangeColor());
    }

    else if (Keyboard.current.rightArrowKey.isPressed)
    {
        rb.linearVelocity = new Vector3(speed, rb.linearVelocity.y, 0);

        StartCoroutine(ChangeColor());
    }

    else
    {
        rb.linearVelocity = new Vector3(0, rb.linearVelocity.y, 0);
    }
}

```

```

IEnumerator ChangeColor()
{
    rend.material = moveMaterial;

    yield return new WaitForSeconds(0.5f);

    rend.material = normalMaterial;
}

```